



Detection of post-COVID-19-related pulmonary diseases in X-ray images using Vision Transformer-based neural network

Anzhelika Mezina^{*}, Radim Burget

Brno University of Technology, FEEC, Department of Telecommunications, Technická 12, Brno, 616 00, Czech Republic

ARTICLE INFO

Keywords:

Image classification
Deep learning
Chest X-ray images
InceptionV3
Vision transformer

ABSTRACT

Objective: Computer methods related to the diagnosis of COVID-19 disease have progressed significantly in recent years. Chest X-ray analysis supported by artificial intelligence is one of the most important parts of the diagnosis. Unfortunately, there is no digital tool dedicated to post-acute pulmonary changes related to COVID-19 and modern diagnostic tools are needed.

Methods: This paper introduces a novel neural network architecture for chest X-ray analysis, which consists of two parts. The first is an Inception architecture that captures global features, and the second is a combination of Inception modules and a Vision Transformer network to analyze the local features. Considering that several diseases can occur in X-ray images together, a specific loss function for multilabel classification was applied — asymmetric loss function (ASL), which we modified for our purpose. In contrast to other works, we focus only on the subgroup of 9 diseases from the chestX-ray14 dataset, which can appear as a consequence of COVID-19.

Results: This work proves the effectiveness of the proposed neural network architecture combined with the asymmetric loss function on post-COVID-related diseases. The results were compared with several well-known classification architectures, such as VGG19, DenseNet121, EfficientNetB4, InceptionV3 and ResNet101. According to the results, the proposed method outperforms the mentioned models with AUC – 0.819, accuracy – 0.736, sensitivity – 0.7683, and specificity – 0.7221.

Significance: Our work is the first one, which focuses on the diagnosis of post-COVID-19 related pulmonary diseases from X-ray images that uses deep learning. The proposed neural network reaches better accuracy than existing well-known architectures.

1. Introduction

Diagnosis of COVID-19 disease has progressed significantly in the last years and months. If at the beginning of the pandemic COVID-19 people relied on preventive endeavours, such as frequent hand washing [1] or wearing facemasks [2], because of a lack of methodology of treatment and overcrowded hospitals, nowadays there are plenty of novel methods and computer-supported systems that make the treatment and care system significantly more effective. Most of the methods focus on the acute phase of the disease, however, the long-term consequences of the disease are nowadays also relatively known. Unfortunately, digital tools that would support health care for the post-COVID-19 phase in many areas are missing. This paper focuses primarily on the diagnosis of pulmonary diseases in the post-acute phase of COVID-19.

One of the most widely used examinations in clinical medicine for post-COVID-19 treatment is a blood test, pulmonary function test and chest imaging using X-ray or computed tomography (CT). Here, the X-ray is often preferred over CT due to its speed, cost and significantly

lower X-ray doses to patients. A chest X-ray image can provide a wide range of information and allows us to check the lungs' condition, evaluate the size of the lungs and heart, follow postoperative changes, monitor the progress and effectiveness of the treatment etc. With that, the technologies for medical image processing, which can assist and help in establishing a diagnosis, are also developing fast. Moreover, with the growing amount of medical images, it is important to process them fast and, more importantly, on time. Because of that, researchers actively study and propose various methods for processing such images, including methods based on machine learning. Some works have already reported post-acute abnormalities [3], and some diseases can even be developed in future, for example, fibrosis [4].

Although the post-COVID-19 disease changes in the lungs are relatively well-known, there are still missing methods specialized in pulmonary disease detection in the post-acute phase of COVID-19. During the pandemic COVID-19 complications with the lungs are getting more critical and it is necessary to pay attention to the possible consequences of COVID-19 in acute and post-acute phases.

^{*} Corresponding author.

E-mail addresses: xmezin00@vutbr.cz (A. Mezina), burgetrm@vut.cz (R. Burget).

<https://doi.org/10.1016/j.bspc.2023.105380>

Received 11 August 2022; Received in revised form 6 July 2023; Accepted 14 August 2023

Available online 25 August 2023

1746-8094/© 2023 Elsevier Ltd. All rights reserved.

The main goal of this work is to propose a neural network architecture that will improve the detection accuracy of post-COVID-19 complications from X-ray images. For this purpose, we propose a specialized neural network architecture and limit it only to long-term COVID-19-related consequences.

The well-known dataset, ChestX-ray14 [5], was utilized for our experiments. Addressing the problem of possible complications in the post-acute phase of COVID-19 we selected a subgroup of 9 diseases from this dataset. To prove the effectiveness of our method, it was compared with selected architectures that are well-known and frequently used in the medical area for the classification tasks: VGG19, DenseNet121, EfficientNetB4, InceptionV3 and ResNet101. The proposed model provides more accurate results in comparison with the mentioned architectures.

The main contributions of this paper are:

- A new model of a neural network based on the transformer and Inception model for X-ray image classification was proposed, which outperforms the well-known modern classification architectures.
- The proposed architecture was optimized for the classification of COVID-19-related diseases.
- Attention was drawn to the possible pulmonary complications of the post-COVID-19 phase.
- The problem of the multilabel and imbalanced dataset was taken into consideration and some modifications in the asymmetric loss function were performed, where it was proved that it has a positive impact on the accuracy.

The rest of this paper is structured as follows: the approaches related to the classification task on ChestX-ray14 and the application of transformers in image processing are presented in Section 2. Section 3 describes the dataset preparation, baseline and the proposed model. After that, the evaluation in Section 4 presents the achieved results. These results are then discussed in Section 5. The last Section 6 concludes the paper.

2. Related works

2.1. Deep learning in pulmonary diseases classification

In recent years, there were a lot of approaches, which aimed to analyze X-ray images using deep learning methods. Last two years, most works are focused on the detection of COVID-19 and pneumonia, however, there were not many works, which are focused on the detection of other pulmonary diseases. They are based on the ChestX-ray14 dataset since it is large and well-processed for machine learning tasks.

For example, an approach in [6] utilizes a U-Net neural network for lung segmentation and uses two branches: local and global to capture modalities of the chest X-ray images. Additionally, the authors apply a self-adaptive weighted fusion scheme for self-tuning the importance of different modalities.

Another work [7] applied the so-called dual asymmetric subnetworks based on DenseNet and ResNet to find more complementary features from the input image for the multilabel classification task. In [8] the authors proposed an architecture, which contains two branches of ResNet-50: local and global, and a fusion branch. However, the model with two branches can perform not only the extraction of global and local features but also other tasks. For example, work [9] proposed to use one of the branches to extract critical disease-specific features and the second one is a feature integrator, which is used to enhance the latent semantic dependencies in the feature space. Also, there are methods, which are based on an ensemble of some networks. One of them is described in [10]. The architecture consists of five DenseNet-169 networks and has the average layer to perform the output of the model. The DenseNet was also used as a feature extractor in [11]. After feature extraction by DenseNet, Logistic Regression is used to perform the classification task.

As can be noticed, most works utilize the DenseNet [12] or ResNet [13] as a feature extractor. In this work, the Inception modules and network [14] are applied to perform the feature extraction.

2.2. Transformers for image classification

One of the recent interesting fields of research is the application of the so-called transformers to image processing, especially for classification tasks. One of the standout works is Vision Transformer (ViT), which was presented by A. Dosovitskiy et al. [15]. ViT uses the same principle as transformers for text processing: the image is divided into small areas, so-called patches, and processed by transformer layers. After that, the output is classified.

There are several approaches for image classification based on ViT with some modifications. For example, work [16] proposes the hierarchical transformer with shifted windows. This principle allows us to process the image at different scales and to use the architecture not only for classification but also for object detection and semantic segmentation. This architecture was also utilized for COVID-19 detection in chest X-ray images [17]. Another modification of ViT is described in [18]. The authors used two branches for multi-scale feature representations in the classification tasks.

Another application of transformers was found in the detection of tuberculosis in X-ray images. The authors used EfficientNet as the backbone, which is followed by a transformer encoder and decoder [19].

Addressing the problem of multilabel classification, the model C-Tran [20] is proposed and it is supposed to be universal for this type of problem. The main difference to other methods is a representation of the state of the labels as positive, negative, or unknown using label masking during training.

Taking into consideration, that ViT has a good potential for application in image classification, especially in X-ray images, it was utilized in this work. Additionally, we are the first, who is focusing on the investigation of possible pulmonary complications post-acute phase of COVID-19 using artificial intelligence (AI).

2.3. Detection of COVID-19 using deep learning

Currently, the detection of COVID-19 from X-ray images is still an active area of research. In most cases, this detection is based on the classification of X-ray images with two classes: COVID-19 and non-COVID-19, or 4 classes: normal, bacterial pneumonia, viral pneumonia and COVID-19 [21]. These methods are based on the application of some traditional machine learning methods [22], or deep learning, primarily architectures based on convolutional neural networks.

For example, work [23] proposes the method, which utilizes object detection with YoloV3, and convolutional neural network (CNN) and histogram of oriented gradients (HOG) for feature extraction. According to the results, it is possible to achieve an accuracy of 99.5%. The object detection model, Single Shot Multibox Detector (SSD), was also used in [24]. The combination of DenseNet101 and SSD allows for achieving a precision of 0.9301.

Another widespread way to perform classification is to utilize the fine-tuning of well-known architectures, such as VGG16, VGG19, DenseNet, InceptionV3, etc. The work [25] provides the comparison of these architectures, which were used for the classification of COVID-19 and pneumonia. The best results are achieved by DenseNet121. It is also possible to use a combination of those models, such as VGG19 and two DenseNet169, and their output can take part in getting the final prediction [26]. Additionally, these architectures can be used not only for X-ray images but also for CT images. Another work [27] provides the comparison of the performance of different models, where the ResNeXt101 has achieved the best accuracy (98%).

There are also approaches, which are based on lung segmentation, as the preprocessing step. For example, work [28] performs the segmentation using ResUNet and the final decision is done with an ensemble

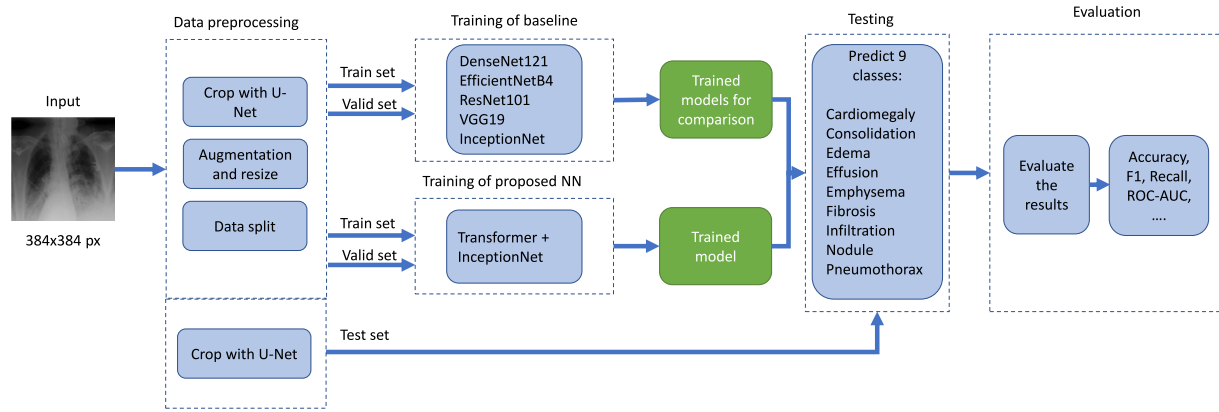


Fig. 1. The overall experiment.

classifier. The final classification can be performed with the traditional machine learning (ML) methods, such as support vector machine (SVM) with a radial basis function (RBF) kernel, XG-boost, and Random Forest [29]. The approach [30] used the same principle: the authors utilized CNN. However, they replaced the last layers with the k -nearest neighbors classifier. Also, CNN can be used in combination with other deep learning models, for example, with Long Short-Time Memory (LSTM) [31] or with CycleGAN, which is used in the preprocessing step [32].

2.4. Summary

According to the studied literature, it can be concluded that most works focus on the diagnosis of the acute phase of COVID-19. However, there is no approach which would focus on the post-acute phase using deep learning. Since those approaches detect different artifacts in the image, they cannot be directly compared. There are some works, which perform statistical evaluation of mortality and readmission because of COVID-19 [33]. However, artificial intelligence was not applied for post-acute image analysis. That is why this work fills the gap in research by pointing out this problem.

3. Methodology

This section presents the description of data processing and the proposed method for the detection of COVID-19-related diseases. The overall scheme of the experiment is depicted in Fig. 1. As can be seen, the first step is dataset preparation and splitting of the data into training and testing sets. The second step is the training of selected neural networks with the training set, and the final part is the evaluation of the model on the testing set and processing of the results. In Section 3.1 the preprocessing of the dataset is described. The next step is the preparation of the baseline (Section 3.2). Here the basic models are introduced, which are used for comparison with the proposed model. After that, a novel model is introduced (Section 3.3).

3.1. Data preprocessing and diseases selection

For the classification of thorax diseases in X-ray images, the large-scale dataset ChestX-ray14 was used. This dataset contains a total of 112,120 images with 14 thorax diseases and is widely used for research purposes.

However, this work focuses on a defined group of diseases, which can appear as a consequence of infection with SARS-CoV-2 (COVID-19). This selection of diseases is based on studies, which were recently reported. Paper [34] provides information about patients, who were positively tested for COVID-19. Its findings are diseases like pericardial effusion, myocarditis and cardiomegaly. Also, paper [35] reports the

statistics on COVID-19-positive patients and their following pulmonary diseases, such as cardiomegaly, pleural effusion and pneumothorax. The relationship between COVID-19 complications and cardiomegaly, pulmonary edema, and pleural effusions was mentioned in [36]. The analysis of X-ray images, which were collected at the beginning of the pandemic showed findings such as nodules, effusion, and cardiomegaly. Among the mentioned diseases, pulmonary fibrosis can also appear. Article [37] provides the details according to this disease, and the work [38] gives the statistical analysis. Also, emphysema and pneumothorax are possible complications of the severe course of COVID-19 [39]. The infiltration was also reported in COVID-19 patients, who, notably, had the tendency to have a headache [40]. The statistical analysis of patients with confirmed COVID-19 is provided in work [41]. Additionally, the pulmonary nodule can also appear as a consequence of COVID-19 [42,43].

The following diseases were selected according to the studied literature:

1. Cardiomegaly
2. Consolidation
3. Edema
4. Effusion
5. Emphysema
6. Fibrosis
7. Infiltration
8. Nodule
9. Pneumothorax

The next step of preprocessing is to get the area of interest in the input image. This was achieved with the segmentation of lungs by a pre-trained U-Net model and followed by cropping. It allows the models to focus on the lungs and exclude unnecessary areas. This step makes the training process more efficient. The example of a cropped image is shown in Fig. 3. After cropping, the image is resized to 384×384 px. To avoid overfitting, the augmentation was also done with the following parameters: height shift range: 0.1, width shift range: 0.1, rotation range: 10, and zoom range: 0.2.

For this experiment, the selected dataset was split into training and testing subsets using the official split of the dataset, which makes this experiment reproducible. Additionally, the validation subset is used in the training process, which is 20% of the training subset. The total number of images in the training subset is 23,496, the validation subset – 5874, and the testing subset – 13,871. The information regarding the number of samples related to each disease is shown in Fig. 2.

3.2. Baseline

To compare the proposed model with other methods, well-known models are selected for the training and testing phases. These include

Table 1
Parameters in inception modules.

Layer	Filters (module 1)	Filters (module 2)	Filters (module 3)	Kernel (for all modules)	Stride (for all modules)
Layer 1	32	16	8	1	2
Layer 2	48	24	12	1	1
Layer 3	64	32	16	3	2
Layer 4	8	4	2	1	1
Layer 5	16	8	4	5	2
Layer 6	16	8	4	1	2

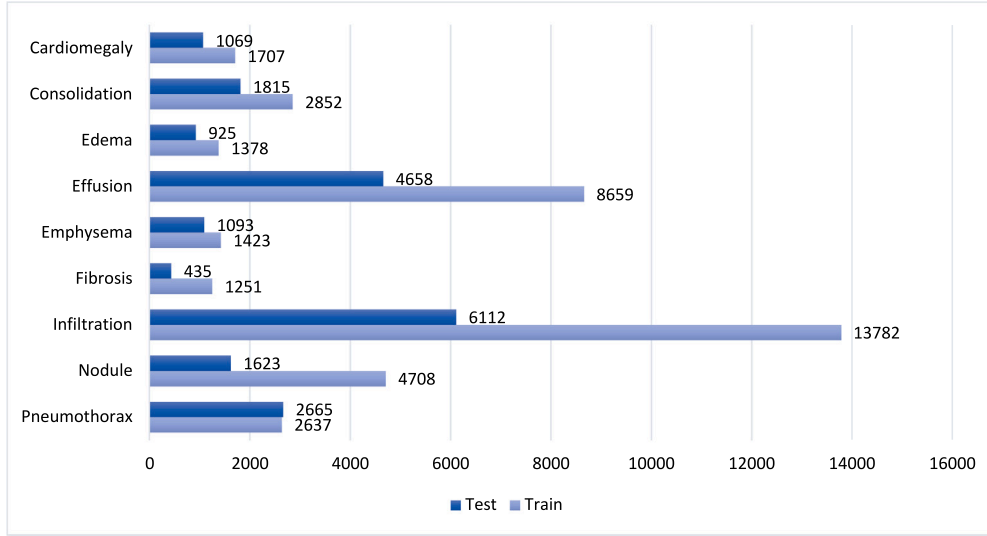


Fig. 2. Diseases distribution in training and testing datasets.

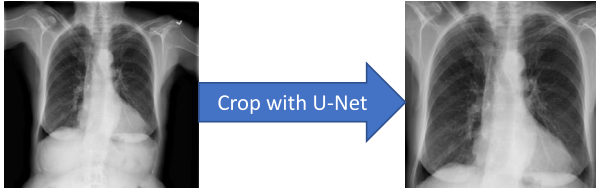


Fig. 3. Preprocessing step - cropping image.

DenseNet121 [12], EfficientNetB4 [44], InceptionV3 [14], ResNet101 [13], and VGG19 [45].

Since the images in the dataset contain one or more diseases, this is a so-called multilabel classification problem. For this purpose, the last layer with softmax activation was removed. Instead, the three new dense layers were added with 64, 32, and 9 neurons. The sigmoid activation function was applied in the last layer for all mentioned methods. The sigmoid activation function [46] gives the output in the range 0 to 1 and can be defined as Eq. (1):

$$f(x) = \frac{1}{1 + e^{-x}} \quad (1)$$

All models were trained with AdamW optimizer [47] with a learning rate of 0.0001, weight decay of 0.00001, and binary cross-entropy (BCE) loss function. The training hyperparameters are the following: batch size 8, number of steps per epoch 1000, and number of epochs 100.

3.3. Proposed architecture

In this work, a new neural network architecture is introduced. It is designed for the multilabel classification task and consists of two branches for feature extraction: the first one is aimed to capture the

global features from the input image, and the second one is used to process the local features and small areas. This architecture is depicted in Fig. 4.

3.3.1. Global branch

The global branch consists of the InceptionV3 model and some additional layers, such as Global Max Pooling, Dense and Dropout. This model is used as a backbone because the nature of the Inception model allows to extract features from bigger or smaller areas. This principle is suitable for our type of problem: the abnormalities in the lungs do not have a fixed size and can be found in a small or big area.

3.3.2. Local branch

The local branch has a more complex structure. Firstly, the feature extraction is performed with three Inception modules with dimension reductions [48]. This stage plays an important role in image processing because it allows to define the area of interest [49]. During these operations, the size of the input image is reduced by 8 times and the output tensor has a size of $48 \times 48 \times 32$. It has some advantages: it reduces the dimensions of input data and, consequently, decreases the training time and memory usage. Moreover, taking into consideration the behavior of the inception module, it also captures areas with different scales from input images thanks to convolutional layers with different kernel sizes. The scheme of the inception module is shown in Fig. 5 and the detailed information about layers' parameters is introduced in Table 1. The names of layers in inception modules presented in Fig. 5 correspond with used names in Table 1. In this way, the details with different sizes are extracted by the model. To visualize the work of these modules, the generated heatmaps of the output of the last inception module are depicted in Fig. 6. As can be seen, the inception modules focus on areas with lungs.

The second part of the local branch is the ViT. ViT was applied in classification tasks and has achieved promising results. Some works were already described in Section 2.2. Fig. 7 explains the details of

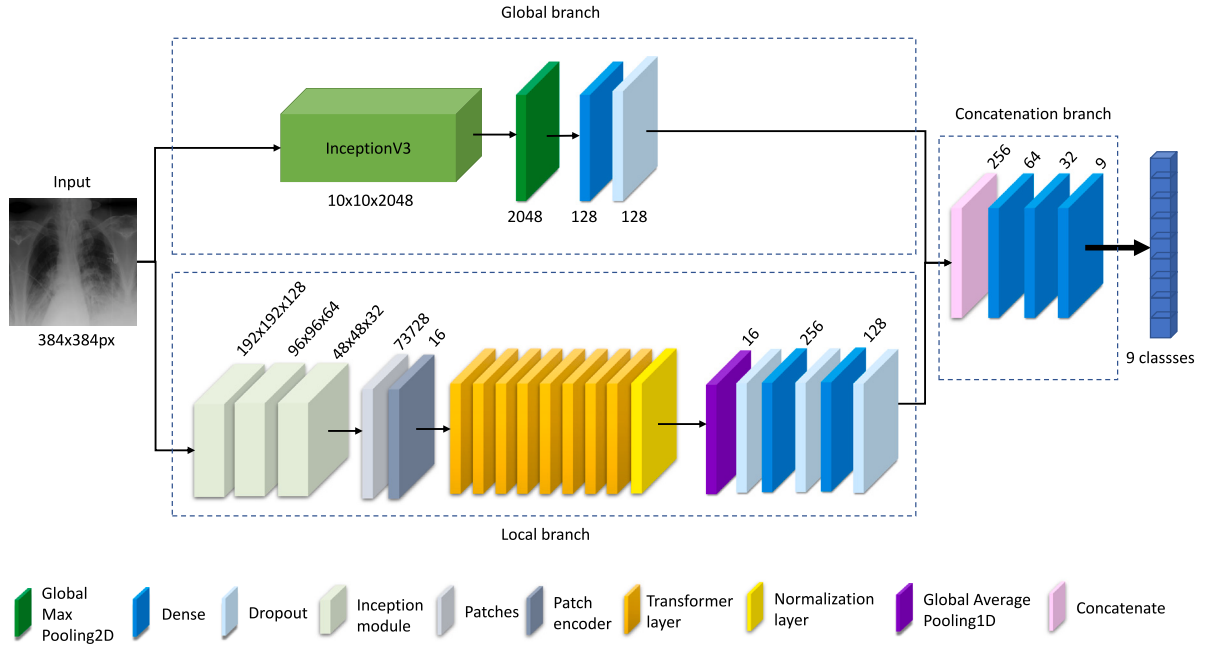


Fig. 4. Proposed neural network architecture.

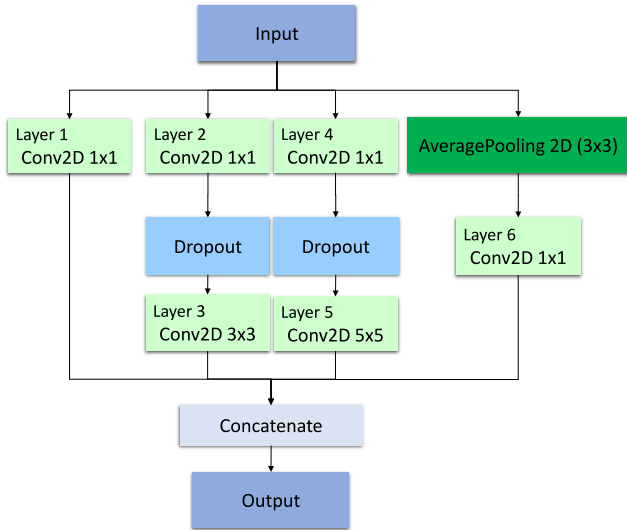


Fig. 5. Inception module.

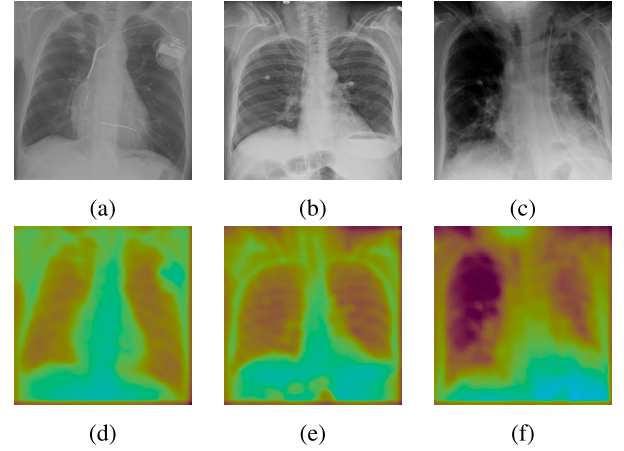


Fig. 6. Input images (a) – (c) and corresponding generated heatmaps (d) – (f).

the ViT architecture. After feature extraction by Inception modules, the output is encoded with a projection dimension with size 16. After that, 8 transformer layers follow. The structure of the transformer layer is left as the original one, and its scheme is shown in Fig. 7. As can be seen, first, the data go through the normalization layer, a multi-head attention layer with 32 attention heads, and they are added to the input of the transformer layer. Then the output of add operation is performed with layer normalization, 2 dense and dropout layers. Dense layers have 32 and 16 units.

The core of the transformer model is the application of multi-head attention [50]. The main idea of the attention function is to map the query and set of key-value pairs to the output. The input of the attention function consists of Q – set of queries, K – keys and V – values.

Firstly, the dot products of a query are computed with all keys. After that, they are divided by $\sqrt{d_k}$, where d_k is the dimension of queries and

keys. And, finally, the softmax function is applied to obtain the weights of the values. Mathematically, this function is defined as Eq. (2):

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V. \quad (2)$$

The final part of this function is the multi-head attention function. It allows the model to process information from different representation subspaces at different positions [50]. In this step, the attention function is performed in parallel for every input and the output is a sequence of vectors. The function is as Eq. (3):

$$MultiHead(Q, K, V) = concat(head_1, \dots, head_h)W^O, \quad (3)$$

where $head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$. Here $W_i^Q \in \mathbb{R}^{d_{model} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{model} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{model} \times d_v}$, $W^O \in \mathbb{R}^{hd_v \times d_{model}}$.

Further, the representation module processes the output of the block with transformer layers (see Fig. 7). This part consists of Layer Normalization, Global Average Pooling and Dropout layers. And the final stage is a Multi-layer perceptron, which consists of two Dense layers with 256 and 128 neurons and a Gaussian Error Linear Unit (GELU) activation function, and a Dropout layer with a rate of 0.5.

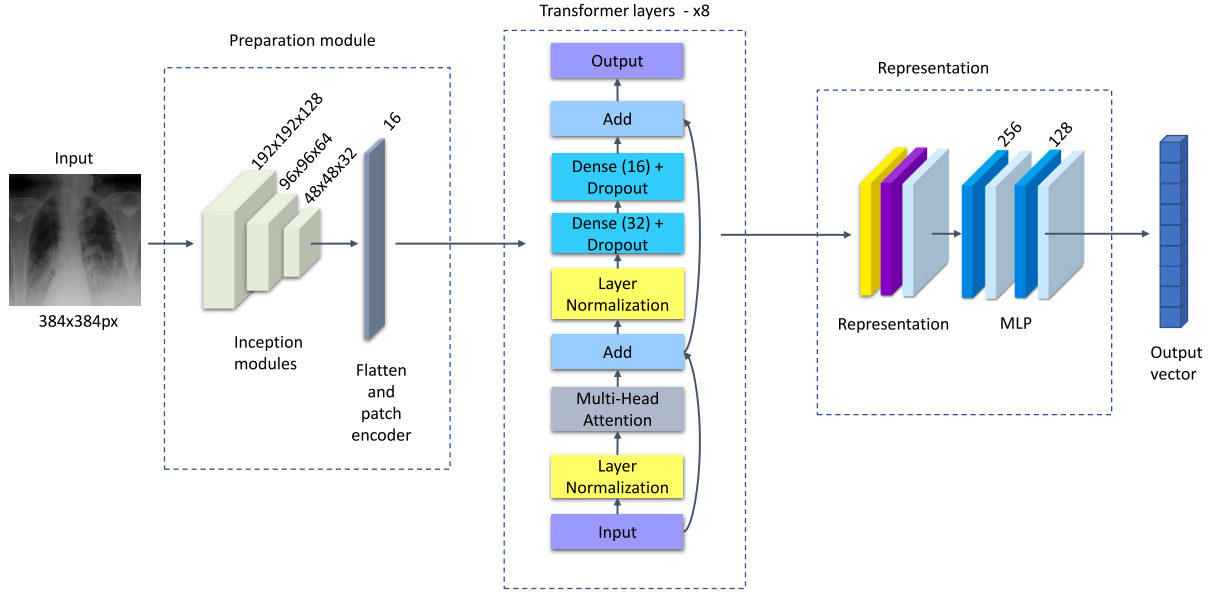


Fig. 7. Transformer network.

3.3.3. Concatenation branch

The last part is the concatenation of global and local branches, which is performed by 3 Dense layers with 64, 32, and 9 units, where the last one has a sigmoid activation function.

3.3.4. Used hyperparameters

Used hyperparameters are the same as the models from the baseline: batch size 8, the number of steps per epoch 1000, and the number of epochs 100. The used optimizer is AdamW with a learning rate of 0.0001 and weight decay of 0.00001.

The used loss function is Asymmetric Loss (ASL) for Multi-Label Classification [51]. However, the focal loss function was utilized instead of applying cross-entropy as a base for the loss function. It allows to achieve better results because the focal loss is aimed to process imbalanced datasets. The asymmetric loss function is described as Eq. (4) [51]:

$$ASL = \begin{cases} L_+ = (1 - p)^{\gamma_+} \log(p) \\ L_- = (p_m)^{\gamma_-} \log(1 - p_m) \end{cases} \quad (4)$$

where L_+ and L_- are the positive and negative loss parts, p is an output probability from the network, p_m is a shifted probability, γ is a focusing parameter. The set up parameters are $\gamma_- = 5$, $\gamma_+ = 1$, clip 0.001.

The focal loss function is defined as Eq. (5) [52]:

$$FL(p_i) = -\alpha_i (1 - p_i)^\beta \log(p_i), \quad (5)$$

where p_i is the model's estimated probability for the class with label $y = 1$, α_i is the balancing factor, and β is the modulating factor.

To show the effectiveness of the asymmetric loss function, the experiment with the application of binary cross-entropy was also performed. The binary cross entropy loss function is defined as Eq. (6) [53]:

$$BCE = \sum_{i=1}^C \hat{y}_i \log(y_i) + (1 - \hat{y}_i) \log(1 - y_i), \quad (6)$$

where C is the total number of classes, y_i is the prediction for the i th class, $\hat{y}_i$ is the desired output.

4. Results and evaluation

4.1. Metrics

The metrics used are accuracy, precision, recall and F-score, which are usually used in the evaluation of classification tasks. The definitions of the mentioned metrics are introduced below (Eq. (7)–(11)) [54]:

Accuracy describes how the trained model is right.

$$Accuracy = \frac{TN + TP}{TP + TN + FP + FN}, \quad (7)$$

where TN – True Negatives, TP – True Positives, FP – False Positives, FN – False Negatives.

Precision is a ratio of right positive predictions over all predicted labels, which are determined as positive.

$$Precision = \frac{TP}{TP + FP} \quad (8)$$

Sensitivity (recall) is a ratio of right positive predictions over the total number of all positive samples.

$$Sensitivity = \frac{TP}{TP + FN} \quad (9)$$

Specificity is a ratio of right negative predictions over the total number of all negative samples.

$$Specificity = \frac{TN}{TN + FP} \quad (10)$$

F1 score is a harmonic mean of precision and recall, which is used to measure the model trained from an imbalanced dataset.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (11)$$

4.2. Results

According to results shown in Table 2, the proposed method outperforms the baseline: all classes, except emphysema and infiltration, have achieved higher values. For comparison, the proposed model with the BCE loss function and the InceptionV3 were also tested. Comparing the results of the proposed model (BCE) and InceptionV3 in Table 2, it can be seen, that almost every class, except fibrosis, has a higher ROC-AUC value and the difference between them is 1%–4%. The fibrosis disease

Table 2
ROC-AUC for all methods.

Pathology	DenseNet121	EfficientNetB4	ResNet101	VGG19	InceptionV3	Proposed (BCE)	Proposed (ASL)
Cardiomegaly	0.876	0.865	0.873	0.852	0.872	0.8854	0.8875
Consolidation	0.665	0.646	0.660	0.633	0.659	0.6659	0.6812
Edema	0.815	0.782	0.806	0.758	0.815	0.8213	0.8230
Effusion	0.804	0.799	0.798	0.764	0.795	0.8126	0.8126
Emphysema	0.897	0.873	0.874	0.725	0.857	0.8963	0.8955
Fibrosis	0.827	0.824	0.823	0.717	0.837	0.8350	0.8540
Infiltration	0.698	0.683	0.689	0.638	0.680	0.6957	0.6966
Nodule	0.776	0.760	0.765	0.649	0.757	0.7760	0.7829
Pneumothorax	0.862	0.856	0.857	0.799	0.859	0.8737	0.8737

Table 3
Results for all methods.

Method	AUC	Sensitivity	Specificity	Accuracy	Precision	Recall	F1 Score
DenseNet121	0.8023	0.7633	0.7131	0.7211	0.3165	0.7624	0.4170
EfficientNetB4	0.7875	0.7255	0.7194	0.7254	0.3076	0.7247	0.4073
ResNet101	0.7936	0.7345	0.7215	0.7249	0.3114	0.7337	0.4087
VGG19	0.7263	0.7042	0.6501	0.6575	0.2649	0.7035	0.3547
InceptionV3	0.7924	0.7235	0.7334	0.7328	0.3178	0.7228	0.4127
Proposed (BCE)	0.8069	0.7506	0.7293	0.7360	0.3216	0.7497	0.4245
Proposed (ASL)	0.8119	0.7683	0.7221	0.7321	0.3219	0.7675	0.4260

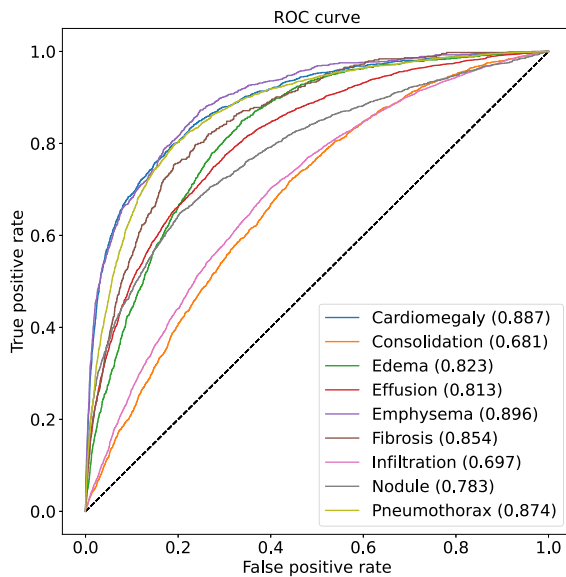


Fig. 8. ROC-AUC for the proposed neural network.

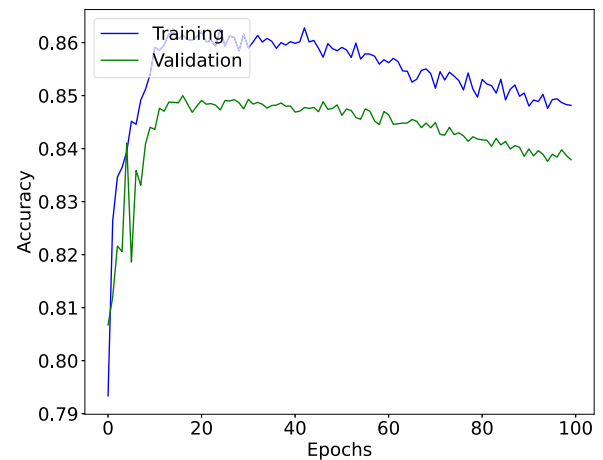


Fig. 9. Accuracy during training of the proposed neural network.

has a difference of 0.002. This comparison proves the positive effect of the application of the local branch with a ViT.

Also, [Table 3](#) with other metrics shows that the proposed model, even with BCE loss function, provides better results, except for specificity.

[Table 2](#) shows that the proposed model has achieved the best results almost for all the diseases, except emphysema and infiltration. The ROC-AUC curve is also depicted in [Fig. 8](#). For these diseases, the DenseNet121 is better with the differences of 0.0014–0.0015. Taking into consideration that these differences are relatively small, it can still be considered that the proposed model is competitive.

Most results from [Table 2](#) show that the proposed model with the application of ASL is better than with the application of BCE as a loss function. Just in the case of effusion and pneumothorax, the results are comparable. This fact is proof of the positive effect of the ASL loss function. The other metrics, presented in [Table 3](#), also provide the information that our architecture, in combination with ASL, achieves better results, except for specificity and accuracy, which are close to the best values. In spite of that, the difference between accuracy values is

also minimal, and possibly its reason is only the initial random values for the training process. The best results are AUC of 0.8119, sensitivity of 0.7683, precision of 0.3219, recall of 0.7675, F1 score of 0.426 by the proposed method with ASL, specificity of 0.7334 by InceptionV3, accuracy of 0.736 by the proposed method with BCE.

VGG19 has the worst results: AUC of 0.7263, sensitivity of 0.7042, specificity of 0.6501, accuracy of 0.6575, precision of 0.2649, recall of 0.7035, F1 score of 0.3547. More detailed results of the proposed model (ASL) per each disease are provided in [Table 4](#).

Cardiomegaly disease has the highest specificity (0.8167) and accuracy (0.8149). However, the best results of AUC and sensitivity have been achieved for emphysema. Generally, all diseases have achieved the AUC in the range of 0.68–0.89, sensitivity 0.64–0.87, specificity 0.55–0.81, accuracy 0.57–0.81, precision 0.11–0.58, recall 0.64–0.87, F1 score 0.19–0.65.

The training process is shown in [Fig. 9](#). From the graph, it can be concluded that the model was trained during the first 20 epochs, and after that, the accuracy slowly decreased.

Additionally, the confusion matrices are shown in [Fig. 10](#). As it was mentioned before, the solved problem is multilabel classification, where one image can be related to several classes (diseases). That is why confusion matrices are computed independently for each class, as a more suitable form of representation in this case. The label “0”

Table 4
Detailed results of proposed neural network (ASL).

Pathology	AUC	Sensitivity	Specificity	Accuracy	Precision	Recall	F1 Score
Cardiomegaly	0.8875	0.7930	0.8167	0.8149	0.2653	0.7923	0.3975
Consolidation	0.6813	0.7240	0.5537	0.5759	0.1962	0.7234	0.3087
Edema	0.8231	0.8490	0.6632	0.6755	0.1524	0.8476	0.2584
Effusion	0.8127	0.7750	0.6984	0.7241	0.5651	0.7748	0.6535
Emphysema	0.8956	0.8730	0.7518	0.7613	0.2311	0.8719	0.3653
Fibrosis	0.8540	0.7540	0.8094	0.8076	0.1132	0.7517	0.1968
Infiltration	0.6966	0.7010	0.6018	0.6453	0.5808	0.7005	0.6351
Nodule	0.7830	0.6480	0.7967	0.7792	0.2968	0.6476	0.4070
Pneumothorax	0.8738	0.7980	0.8076	0.8058	0.4966	0.7981	0.6123

indicates the disease is not present in the image, and “1” stands for positivity regarding the disease’s presence. The main diagonal represents how many samples were correctly classified, or, in other words, if the model is able to detect the disease. These results correspond with the accuracy introduced in Table 4. Fig. 10 shows that the model detects better such diseases as effusion (26%) and infiltration (31%). Also, the detection of pneumothorax can be considered as successful (15% correctly predicted). On the other hand, cardiomegaly has high accuracy (81%), but the TP is 6% and TN is 75%. Similar results are for fibrosis: TP is 2% and TN is 78%, and pneumothorax: TP is 15% and TN is 65%. Some other diseases have relatively higher false positive results: consolidation – 39%, edema – 31%, emphysema – 23%, fibrosis – 18%, nodule – 18%. These confusion matrices confirm the results for precision, which were presented in Table 3. On the other hand, there are fewer values predicted as a false negative: cardiomegaly – 2%, consolidation – 4%, edema – 1%, effusion – 8%, emphysema – 1%, fibrosis – 1%, infiltration – 13%, nodule – 4%, pneumothorax – 4%, so it can be concluded that the model with less probability will predict the disease falsely negative.

5. Discussion

According to the achieved results, the proposed method works more effectively than other well-known architectures, such as DenseNet121, EfficientNetB4, ResNet101, VGG19, and InceptionV3. From the introduced results in the previous subsection, the worst results are obtained by VGG19. The possible reason is the simplicity of the model. On the other hand, more complex architectures, such as DenseNet121, ResNet101 or InceptionV3 are able to achieve better results than VGG19. Also, DenseNet121 has higher ROC-AUC for diseases like emphysema and infiltration. InceptionV3 has worse results compared to our proposed model, but better than other models. Also, DenseNet121 has achieved competitive results with the proposed model.

Here we proved that a combination of InceptionV3 and Vision transformer can give more accurate results than using only InceptionV3. This fact is supported by results for both cases: training and testing were performed for InceptionV3 and the proposed model. The next point is that utilization of the ASL function allows the model to train better and give accurate results. To prove it, the comparison of the model trained with traditional binary cross-entropy and trained with ASL was provided.

According to the confusion matrices, there is a low rate of false negative predictions, which is very important in medicine: the disease with low probability will be missed. However, there is a tendency that the number of FP to be larger than the number of TP predictions. Here can be noted, that it is a less crucial problem, than a large number of FN, because it is better to suspect some disease and pay attention to it than it will be missed and, consequently, lead to worse complications.

The model gives more accurate predictions in terms of TP and TN for the diseases, which have more X-ray images (see Fig. 2), such as effusion and infiltration. The third highest number of images per class is pneumothorax and the confusion matrix for this class shows also relatively good results. The main reason for this misclassification is an imbalance of the dataset (which is obvious from class distribution) and

the relatively low number of samples for other diseases. Here can be concluded, that there is a necessity to extend the dataset by adding X-rays with these other classes, or to apply the advanced techniques of preprocessing to deal with such a problem.

On the other hand, the results introduced with confusion matrices show, which classes can be complicated to detect and which of them should be considered for more detailed investigation.

The closest problem to our topic is the detection of pulmonary diseases, which was discussed in Section 2.1. However, these works conducted experiments over the whole dataset, but not over part of it with selected labels.

Compared with work [6], which has achieved for cardiomegaly AUC value of 0.87, and fibrosis – 0.833, our approach has AUC of 0.875, and 0.854 respectively. Another work [7] has achieved AUC for fibrosis 0.837. The work [9] has results for fibrosis – 0.831 and infiltration – 0.699, nodule – 0.775, pneumothorax – 0.871, but our model performed with results 0.854, 0.6966, 0.7829, and 0.8737 respectively. Our model also outperformed the approach described in [10] for disease effusion (0.8037 vs 0.8126), fibrosis (0.8187 vs 0.8540), infiltration (0.6961 vs 0.6966), nodule (0.7534 vs 0.7829), and pneumothorax (0.8515 vs 0.8737). The provided results show, that other models have worse detection capability for fibrosis disease than our model.

Another point is, the used image size in the mentioned works is different from ours: for example, they used size 256×256 px, 512×512 px or 1024×1024 px. With increasing the dimension size, it is getting more time-consuming and memory expensive for training and prediction phases. For that reason, we used a lower size, 384×384 px, but preserved the details in images. In spite of a large number of works, which have focused on pulmonary diseases, there is no other work which would be aimed to classify the limited group of post-COVID-19 pulmonary diseases. Because of that reason, the comparison was performed with well-known classifiers to provide fair and objective results.

It is worth noting that many works utilize architectures, such as DenseNet, ResNet, VGG19, etc. However, our approach is based on the utilization of InceptionV3 architecture and Transformer, which are not often used in this field of research.

Another popular topic in this field of research is the detection of COVID-19 in the acute phase. The main limitation of such works is their application in the real world. Last year there are much less complications during the acute phase of COVID-19 and there is a risk, that approaches for the acute phase will fail. However, patients still suffer from other pulmonary diseases in the post-acute phase. This work is the first one, which points out the problem of the post-acute phase of COVID-19.

The main limitation of this work is the lack of real data on post-COVID-19 patients. Unfortunately, there is no such dataset available, and since a large dataset would be necessary, it is unrealistic that such a database will be created in the coming years. So, the only possibility, at least now, is to train and test on the existing datasets with pulmonary diseases. In our case, the dataset ChestX-ray14 was used. On the other hand, it is the task for future work: collecting real post-COVID-19 data and testing with them the proposed architecture. However, it is

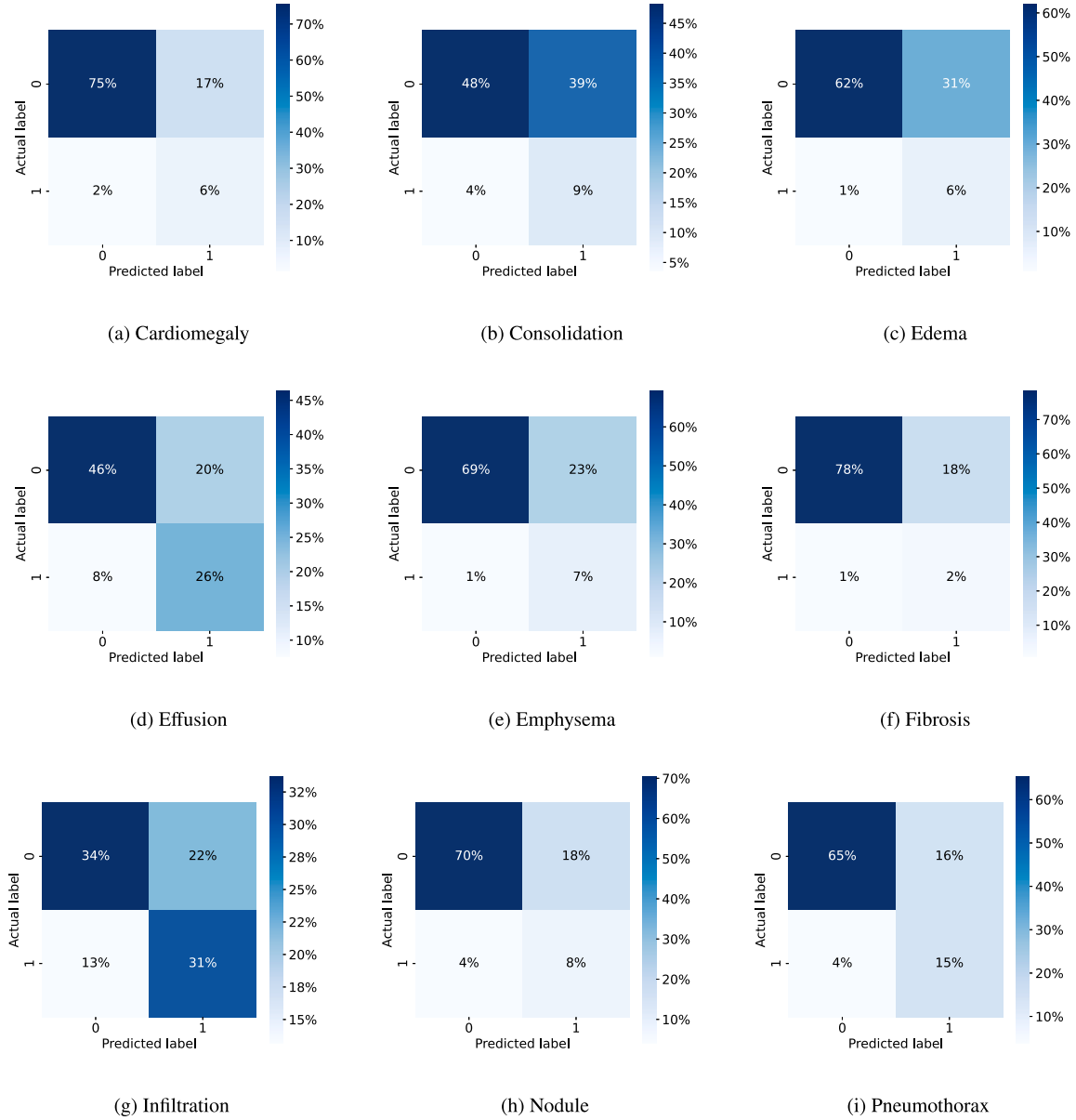


Fig. 10. Confusion matrices for each class.

worth paying attention to the classes which obtained worse results than others: infiltration and consolidation.

The results obtained with the proposed model are promising, it performs well. However, the Inception model and inception blocks are relatively computationally demanding during training because of their complex structure. It is a possible issue to be solved in future work. Another point is that the ViT has some parameters, such as the number of layers, dimension projection, etc., which have the potential for possible further optimization. The research for an optimal combination of hyper-parameters is not an easy task, and this problem should be studied.

6. Conclusion and future directions

This work focuses on the detection of post-COVID-19 diseases from chest X-ray images. We introduced a novel neural network architecture composed of local and global branches that focus on nine COVID-19-related pulmonary diseases (Cardiomegaly, Consolidation, Edema,

Effusion, Emphysema, Fibrosis, Infiltration, Nodule and Pneumothorax). This model is based on the Inception model and ViT and we proved that the ASL loss function improves the accuracy of the resulting model.

The proposed architecture was compared with other well-known and frequently used architectures for classification, such as DenseNet121, EfficientNetB4, InceptionV3, VGG19, and ResNet101. According to the results, our method has achieved an accuracy of 0.7321, AUC of 0.8119, sensitivity of 0.7683, specificity of 0.7221, precision of 0.3219, recall of 0.7675, and F1-score of 0.426, which outperformed the other architectures. The provided results prove that the application of a local branch with inception modules and a ViT can make the model more accurate in predictions.

One of the limitations of this work is the lack of previous works, which would focus on the group of some diseases. Consequently, the proposed model was compared with some well-known architectures. The next thing is the lack of real post-COVID-19 data, which is why the well-known dataset was used for our purpose. However, it can

be considered, that the proposed model can be still effective over real-world data.

The next point is the possible low rate of F1 score and precision. In spite of high ROC-AUC values and accuracy results, the other metrics should be also considered, so it would be possible to identify the weaknesses of the proposed approach or which diseases are challenging for detection.

In future work, the mentioned issues should be solved by collecting a dataset of large real-world data. Additionally, the application of ViT in the medical field of research can be improved by a combination with convolutional layers and possibly, the accuracy of disease detection can be increased. Also, it is important to extend the research with experiments with different optimizers and loss functions. As it was shown, it has an impact on results. Another thing which can be done is the fine-tuning of the trained model on ChestXray-14 to the real data.

Funding

This work was supported by the Ministry of the Interior of the Czech Republic, under Grant no. VK01010107 and VK01010153.

CRediT authorship contribution statement

Anzhelika Mezina: Conceptualization, Methodology, Software, Investigation, Visualization, Writing – original draft. **Radim Burget:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] A.G. Yadessa, A.O. Salau, Low cost sensor based hand washing solution for COVID-19 prevention, in: 2021 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies, 3ICT, IEEE, 2021, pp. 93–97.
- [2] T. Natnael, Y. Alemnew, G. Berihun, M. Abebe, A. Andualem, S. Ademe, B. Tegegne, M. Adane, Facemask wearing to prevent COVID-19 transmission and associated factors among taxi drivers in Dessie City and Kombolcha Town, Ethiopia, *PLoS One* 16 (3) (2021) e0247954.
- [3] J.J. Solomon, B. Heyman, J.P. Ko, R. Condos, D.A. Lynch, CT of postacute lung complications of COVID-19, *Radiology* (2021).
- [4] E. Vasarmidi, E. Tsitoura, D.A. Spandidos, N. Tzanakis, K.M. Antoniou, Pulmonary fibrosis in the aftermath of the COVID-19 era, *Exp. Ther. Med.* 20 (3) (2020) 2557–2560.
- [5] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, R. Summers, ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2017, pp. 3462–3471.
- [6] B. Chen, Z. Zhang, J. Lin, Y. Chen, G. Lu, Two-stream collaborative network for multi-label chest X-ray image classification with lung segmentation, *Pattern Recognit. Lett.* 135 (2020) 221–227.
- [7] B. Chen, J. Li, X. Guo, G. Lu, DualCheXNet: dual asymmetric feature learning for thoracic disease classification in chest X-rays, *Biomed. Signal Process. Control* 53 (2019) 101554.
- [8] Q. Guan, Y. Huang, Z. Zhong, Z. Zheng, L. Zheng, Y. Yang, Thorax disease classification with attention guided convolutional neural network, *Pattern Recognit. Lett.* 131 (2020) 38–45.
- [9] Q. Guan, Y. Huang, Y. Luo, P. Liu, M. Xu, Y. Yang, Discriminative feature learning for thorax disease classification in chest X-ray images, *IEEE Trans. Image Process.* 30 (2021) 2476–2487.
- [10] A.M. Capuchino Antioquia, Accurate thoracic disease classification via ensemble networks, in: 2022 the 5th International Conference on Image and Graphics Processing, ICI GP, 2022, pp. 196–201.
- [11] I. Allaoui, M.B. Ahmed, A novel approach for multi-label chest X-ray classification of common thorax diseases, *IEEE Access* 7 (2019) 64279–64288.
- [12] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4700–4708.
- [13] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
- [14] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826.
- [15] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.
- [16] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.
- [17] J. Jiang, S. Lin, COVID-19 detection in chest X-ray images using swin-transformer and transformer in transformer, 2021, arXiv preprint arXiv:2110.08427.
- [18] C.F.R. Chen, Q. Fan, R. Panda, Crossvit: Cross-attention multi-scale vision transformer for image classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 357–366.
- [19] L.T. Duong, N.H. Le, T.B. Tran, V.M. Ngo, P.T. Nguyen, Detection of tuberculosis from chest X-ray images: boosting the performance with vision transformer and transfer learning, *Expert Syst. Appl.* 184 (2021) 115519.
- [20] J. Lanchantin, T. Wang, V. Ordonez, Y. Qi, General multi-label image classification with transformers, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 16478–16488.
- [21] S. Vineth Ligi, S.S. Kundu, R. Kumar, R. Narayanamoorthi, K.W. Lai, S. Dhanalakshmi, Radiological analysis of COVID-19 using computational intelligence: A broad gauge study, *J. Healthc. Eng.* 2022 (2022).
- [22] A.O. Salau, Detection of corona virus disease using a novel machine learning approach, in: 2021 International Conference on Decision Aid Sciences and Application, DASA, IEEE, 2021, pp. 587–590.
- [23] A.M. Ayalew, A.O. Salau, B.T. Abeje, B. Enyew, Detection and classification of COVID-19 disease from X-ray images using convolutional neural networks and histogram of oriented gradients, *Biomed. Signal Process. Control* 74 (2022) 103530.
- [24] S. Mahajan, A. Raina, X.Z. Gao, A.K. Pandit, COVID-19 detection using hybrid deep learning model in chest X-rays images, *Concurr. Comput.: Pract. Exper.* 34 (5) (2022) e6747.
- [25] A. Shazia, T.Z. Xuan, J.H. Chuah, J. Usman, P. Qian, K.W. Lai, A comparative study of multiple neural network for detection of COVID-19 on chest X-ray, *EURASIP J. Adv. Signal Process.* 2021 (1) (2021) 1–16.
- [26] E. Jangam, A.A.D. Barreto, C.S.R. Annavarapu, Automatic detection of COVID-19 from chest CT scan and chest X-rays images using deep learning, transfer learning and stacking, *Appl. Intell.* 52 (2) (2022) 2243–2259.
- [27] S.L. Woan Ching, K.W. Lai, J.H. Chuah, K. Hasikin, A. Khalil, P. Qian, K. Xia, Y. Jiang, Y. Zhang, S. Dhanalakshmi, Multiclass convolution neural network for classification of COVID-19 CT images, *Comput. Intell. Neurosci.* 2022 (2022).
- [28] R. Hertel, R. Benlamri, A deep learning segmentation-classification pipeline for X-ray-based covid-19 diagnosis, *Biomed. Eng. Adv.* (2022) 100041.
- [29] A. Bhattacharyya, D. Bhaik, S. Kumar, P. Thakur, R. Sharma, R.B. Pachori, A deep learning based approach for automatic detection of COVID-19 cases using chest X-ray images, *Biomed. Signal Process. Control* 71 (2022) 103182.
- [30] S.M.J. Jalali, M. Ahmadian, S. Ahmadian, R. Hedjam, A. Khosravi, S. Nahavandi, X-ray image based COVID-19 detection using evolutionary deep learning approach, *Expert Syst. Appl.* 201 (2022) 116942.
- [31] Z. Mousavi, N. Shahini, S. Sheykhivand, S. Mojtahedi, A. Arshadi, COVID-19 detection using chest X-ray images based on a developed deep neural network, *SLAS Technol.* 27 (1) (2022) 63–75.
- [32] J. Janbi, S. Elnazer, Diagnosing COVID-19 from augmented CTX images dataset using CycleGAN and CCGAN, in: Proceedings of Sixth International Congress on Information and Communication Technology, Springer, 2022, pp. 443–455.
- [33] W.K. Loo, K. Hasikin, A. Suhaimi, P.L. Yee, K. Teo, K. Xia, P. Qian, Y. Jiang, Y. Zhang, S. Dhanalakshmi, et al., Systematic review on COVID-19 readmission and risk factors: Future of machine learning in COVID-19 readmission studies, *Front. Public Health* (2022) 1311.
- [34] A.K. Mishra, A. Lal, K.K. Sahu, M. Kranis, J. Sargent, Quantifying and reporting cardiac findings in imaging of COVID-19 patients, *Monaldi Arch. Chest Dis.* 90 (4) (2020).
- [35] D. Cozzi, M. Albanesi, E. Cavigli, C. Moroni, A. Bindi, S. Luvarà, S. Lucarini, S. Busoni, L.N. Mazzoni, V. Miele, Chest X-ray in new coronavirus disease 2019 (COVID-19) infection: findings and correlation with clinical outcome, *La Radiol. Med.* 125 (8) (2020) 730–737.
- [36] A.J. Winant, E. Blumfield, M.C. Liszewski, J. Kurian, A.M. Foust, E.Y. Lee, Thoracic imaging findings of multisystem inflammatory syndrome in children associated with COVID-19: what radiologists need to know now, *Radiol. Cardiothoracic Imaging* 2 (4) (2020) e200346.

- [37] C. Zhang, Z. Wu, J.W. Li, K. Tan, W. Yang, H. Zhao, G.Q. Wang, Discharge may not be the end of treatment: Pay attention to pulmonary fibrosis caused by severe COVID-19, *J. Med. Virol.* 93 (3) (2021) 1378–1386.
- [38] R.M.M. Ali, M.B.I. Ghonimy, Post-COVID-19 pneumonia lung fibrosis: a worrisome sequelae in surviving patients, *Egypt. J. Radiol. Nucl. Med.* 52 (1) (2021) 1–8.
- [39] W. Xu, X. Luo, H. Wang, C. Shen, Y. Song, T. Sun, M. Chen, Pulmonary emphysema, bullae, and pneumothorax in COVID-19 pneumonia, *Radiol. Case Rep.* 16 (5) (2021) 995–998.
- [40] Ö. Karadaş, B. Öztürk, A.R. Sonkaya, B. Taşdelen, A. Özge, H. Bolay, Latent class cluster analysis identified hidden headache phenotypes in COVID-19: impact of pulmonary infiltration and IL-6, *Neurol. Sci.* 42 (5) (2021) 1665–1673.
- [41] Y.C. Lin, Y.L. Lee, C.Y. Cheng, W.P. Tseng, J.L. Wu, C.H. Lin, M.Y. Chung, C.M. Kang, Y.F. Lee, C.P. Chen, et al., Multicenter evaluation of four immunoassays for the performance of early diagnosis of COVID-19 and assessment of antibody responses of patients with pneumonia in Taiwan, *J. Microbiol. Immunol. Infect.* 54 (5) (2021) 816–829.
- [42] D. Somboonviboon, A. Wattanatham, Pulmonary pseudoaneurysm presented as a lung nodule after recovering from COVID-19 pneumonia, *Respirol. Case Rep.* 10 (6) (2022) e0963.
- [43] A. Diaz, D. Bujnowski, P. McMullen, M. Lysandrou, V. Ananthanarayanan, A.N. Husain, R. Freeman, W.T. Vigneswaran, M.K. Ferguson, J.S. Donington, et al., Pulmonary parenchymal changes in COVID-19 survivors, *Ann. Thorac. Surg.* 114 (1) (2022) 301–310.
- [44] M. Tan, Q. Le, Efficientnet: Rethinking model scaling for convolutional neural networks, in: *International Conference on Machine Learning*, PMLR, 2019, pp. 6105–6114.
- [45] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, arXiv preprint [arXiv:1409.1556](https://arxiv.org/abs/1409.1556).
- [46] A. Mezina, R. Burget, Obfuscated malware detection using dilated convolutional network, in: *2022 14th International Congress on Ultra Modern Telecommunications and Control Systems and Workshops, ICUMT, IEEE, 2022*, pp. 110–115.
- [47] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, 2017, arXiv preprint [arXiv:1711.05101](https://arxiv.org/abs/1711.05101).
- [48] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, A. Rabinovich, Going deeper with convolutions, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 1–9.
- [49] A.O. Salau, S. Jain, Feature extraction: a survey of the types, techniques, applications, in: *2019 International Conference on Signal Processing and Communication, ICSC, IEEE, 2019*, pp. 158–164.
- [50] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [51] E. Ben-Baruch, T. Ridnik, N. Zamir, A. Noy, I. Friedman, M. Protter, L. Zelnik-Manor, Asymmetric loss for multi-label classification, 2020, arXiv preprint [arXiv:2009.14119](https://arxiv.org/abs/2009.14119).
- [52] T.Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.
- [53] D. Pham, S. Koesnadi, G. Dovletov, J. Pauli, Unsupervised adversarial domain adaptation for multi-label classification of chest X-ray, in: *2021 IEEE 18th International Symposium on Biomedical Imaging, ISBI, IEEE, 2021*, pp. 1236–1240.
- [54] N. Indumathi, M. Shanmuga Eswari, A.O. Salau, R. Ramalakshmi, R. Revathy, Prediction of COVID-19 outbreak with current substantiation using machine learning algorithms, *Intell. Interact. Multimed. Syst. e-Healthc. Appl.* (2022) 171–190.